

PRANAY PRASOON

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EDUCATION

Bennett University, Greater Noida

2023 – Present

B.Tech in Computer Science

- **Languages** Python, SQL, Golang, Javascript
- **Frameworks & Libraries** FastAPI, Streamlit, Next.js, Node.js, Numpy, Pandas, Scikit-learn, Statsmodels, Prophet, Plotly
- **Data Visualization Tools** PowerBI, Excel, Matplotlib, Seaborn
- **Databases** PostgreSQL, SQLite, MongoDB, Redis

PROJECTS

- **PulseIQ – Real-Time Public Sentiment & Intelligence Platform**

Tech: FastAPI, NLP (BERT, spaCy), Celery, Redis, MongoDB, TimescaleDB, Elasticsearch, Docker

Github

- Architected an **end-to-end, high-throughput real-time data pipeline** to ingest social data (Reddit, Google News), asynchronously trigger NLP inference, and aggregate insights using **Celery + Redis** task queues.
- Designed a **microservices-based NLP inference system** with a dedicated FastAPI AI service to deploy **fine-tuned BERT emotion classifiers** and **spaCy NER**, decoupling heavy ML workloads from ingestion services for scalability.
- **Polyglot Persistence Strategy:** Engineered an optimized storage layer using **MongoDB** as a data lake for raw documents, **TimescaleDB** for sub-second time-series aggregations, and **Elasticsearch** for low-latency full-text search.
- Built **real-time aggregation and analytics endpoints** to expose sentiment trends, entity insights, and temporal metrics with low-latency APIs.
- Containerized and orchestrated **6+ distributed microservices** using **Docker Compose**, managing heavy AI dependencies and ensuring reliable inter-service communication.

- **Distributed File System (P2P Storage)**

Tech: Go, TCP Networking, Custom P2P Protocol, Goroutines, Channels, AES-CTR

Github

- Engineered a **decentralized, content-addressable distributed file storage system** in Go supporting file sharding, replication, and peer discovery.
- Designed a **custom TCP-based P2P transport protocol** with handshake negotiation and binary stream decoding for efficient node-to-node communication.
- Implemented **end-to-end encryption (AES-CTR)** to secure file transfers across untrusted peers.
- Leveraged **Go concurrency (goroutines, channels, mutexes)** to handle parallel peer connections, broadcasting, and high-throughput file streaming without blocking.

- **SafeChat – Real-Time Moderated Chat Platform**

Tech: FastAPI, Python, Socket.IO, Next.js, Redis, PostgreSQL, Docker, JWT

Github

- Designed and implemented a **scalable real-time chat backend** with **JWT-based authentication**, group and private messaging, **presence tracking**, and **typing indicators**.
- Built an **AI-powered real-time moderation pipeline** that classifies every message before broadcast and enforces approval, censorship, or blocking policies.
- Implemented **persistent messaging and delivery guarantees** using **PostgreSQL** and **Redis** for caching and pub/sub.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Data Science Professional | [view](#)